

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Analyse and Practise

Name: _____

Date: _____

LEARNING INTENTION	measure with reasonable precision and represent data to reveal patterns
KEY IDEA	Choose equipment and units that match the variable. Record repeated data consistently, then select a table, graph or model that makes patterns and relationships visible.
WORKED MODEL	Use the cycle: question -> prediction -> safe method -> precise data -> representation -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN

[] apply the inquiry skill [] use evidence [] evaluate quality [] explain a next step

A. NOTICE AND IDENTIFY

1. Choose suitable equipment and unit for length, mass, temperature and time.

2. Record 4.2, 4.20 and 4.200: explain what precision communicates.

3. Identify missing table features: title, variable headings, units, trials.

B. APPLY AND EXPLAIN

4. Calculate a representative result from three trials and retain sensible precision.

5. Select column graph, line graph or labelled model for three data purposes.

6. Describe pattern, trend and relationship without claiming unsupported cause.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Construct a complete table and graph plan for the scenario.

8. Check scale, intervals, labels, units and faithful representation.

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D. APPLY THE SKILL IN CONTEXT

9. Scenario: measure how shadow length changes at set times Results are 12, 15, 15, 18 and 21. Organise and represent them.

10. State two supported observations and one claim the data cannot support.

11. Identify an anomalous result and decide whether to retain, repeat or investigate it.

E. REASON, EVALUATE AND TRANSFER

12. Critique a graph with a truncated axis and unequal intervals.

13. Redesign it ethically and explain how the interpretation changes.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[] I can measure with reasonable precision and represent data to reveal patterns.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult decision was _____ because _____

One correction or improvement I made: _____

My next practice step: _____

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Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. Match instrument/unit to required variable and scale.
2. Trailing zeros may show measurement resolution; do not imply false precision.
3. All listed features are required.
4. Use mean/median only when appropriate and match decimal precision to equipment.
5. Categories=column; continuous change=line; structure/process=model.
6. Describe only visible evidence and avoid causal overreach.
- 7-8. Representation must be complete, proportional and readable.
9. Use ordered table and suitable equal-interval scale.
10. Observations cite values; unsupported claim exceeds variables/sample.
11. Do not delete automatically; check method and repeat if possible.
- 12-13. Restore honest baseline/intervals and explain reduced distortion.

TEACHER CHECK AND NEXT STEP

[] Uses accurate science ideas and evidence.

[] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

More decimal places do not create greater measurement accuracy.

A graph reveals patterns but does not automatically establish cause.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.